

TACOMA NARROWS, WA, USA

WMO: 727938

Lat: 47.268N

Lon: 122.576W

Elev: 89

StdP: 100.26

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 94274

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-2.4	-0.6	-10.2	1.6	0.6	-7.1	2.1	2.0	11.1	8.2	9.9	8.1	2.3	10	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	28.8	17.9	27.0	17.2	25.0	16.5	18.8	27.0	17.9	25.2	17.1	23.6	3.2	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	11.5	20.2	15.1	10.8	19.5	14.3	10.3	18.9	53.3	27.1	50.5	25.2	48.2	23.6	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.7	7.7	6.8	DB	-4.5	32.9	2.2	2.5	-6.0	34.7	-7.3	36.2	-8.5	37.6	-10.1	39.4
			WB	-5.5	20.3	2.4	0.9	-7.2	21.0	-8.6	21.5	-10.0	22.0	-11.7	22.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	5.6	6.1	7.8	9.9	13.2	15.8	18.5	18.5	16.0	11.5	7.6	5.2
	DBStd	5.49	3.00	2.73	2.58	2.59	2.98	2.89	2.61	2.18	2.42	2.25	2.91	2.85
	HDD10.0	607	139	112	77	32	4	0	0	0	0	10	82	151
	HDD18.3	2642	396	344	327	252	161	89	29	25	77	212	322	408
	CDD10.0	1093	2	1	9	29	104	173	262	263	181	57	9	1
	CDD18.3	84	0	0	0	0	3	12	33	29	8	0	0	0
	CDH23.3	774	0	0	0	3	36	119	316	240	61	0	0	0
	CDH26.7	185	0	0	0	0	4	31	91	50	8	0	0	0
(14) Wind	WSAvg	2.7	2.8	3.0	3.1	2.9	2.7	2.7	2.5	2.3	2.4	2.6	2.9	2.8
(15) Precipitation	PrecAvg	1238	183	128	131	87	55	41	19	27	39	126	196	197
	PrecMax	1517	372	320	268	195	114	84	47	86	126	279	464	333
	PrecMin	897	86	4	28	22	6	6	4	1	3	21	42	71
	PrecStd	197	68	65	56	47	28	21	13	25	30	64	94	80
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.6	14.0	19.2	22.8	26.9	30.5	32.4	30.8	27.9	20.9	16.0	13.4
		MCWB	11.7	9.8	11.3	13.6	15.8	17.8	19.8	18.5	16.7	13.8	13.7	11.6
	2%	DB	11.9	12.4	15.5	19.0	23.8	27.1	29.0	28.0	25.1	18.0	14.0	11.4
		MCWB	10.1	9.2	10.0	11.6	14.6	16.9	18.4	17.8	16.5	13.2	12.3	9.9
	5%	DB	10.7	11.1	13.2	16.6	21.3	24.0	26.9	26.1	22.9	16.5	12.7	10.0
		MCWB	9.2	8.4	9.3	10.7	13.8	15.5	17.5	17.2	15.7	12.7	11.2	8.5
	10%	DB	9.5	9.9	12.0	14.3	18.8	21.6	24.5	24.1	21.0	15.1	11.7	8.8
		MCWB	8.1	7.8	8.6	9.5	12.8	14.6	16.6	16.5	14.9	12.1	10.3	7.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.5	12.0	12.6	14.3	16.8	18.8	20.7	19.6	18.4	16.0	14.4	12.1
		MCDB	13.3	12.9	16.5	21.2	24.2	28.4	30.7	28.0	24.5	18.1	15.5	13.2
	2%	WB	10.7	10.2	11.2	12.6	15.4	17.3	19.1	18.6	17.3	14.5	12.6	10.3
		MCDB	11.3	11.4	13.9	17.7	22.5	25.8	27.7	26.4	23.0	16.6	13.7	11.2
	5%	WB	9.4	9.0	10.2	11.4	14.3	16.1	18.0	17.8	16.5	13.5	11.6	8.8
		MCDB	10.2	10.3	12.7	15.2	20.1	22.9	25.6	24.8	21.3	15.4	12.5	9.7
	10%	WB	8.2	8.0	9.1	10.3	13.2	15.1	17.0	17.0	15.6	12.6	10.5	7.8
		MCDB	9.1	9.3	11.4	13.6	18.0	20.9	23.6	23.2	19.7	14.4	11.4	8.6
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	5.4	6.6	7.7	8.8	9.2	10.7	10.3	8.6	6.1	4.8	4.0
		MCDBR	5.4	7.1	9.2	11.4	13.4	13.6	14.5	13.7	12.1	8.4	5.5	4.8
	5% WB	MCWBR	3.7	4.4	4.8	5.5	6.2	5.9	5.8	5.5	5.1	4.6	3.9	3.7
		MCWBR	4.7	5.5	7.9	9.8	11.5	12.1	13.1	12.2	10.2	6.7	5.0	4.5
(40) Clear-Sky Solar Irradiance	taub		0.327	0.314	0.327	0.353	0.375	0.363	0.349	0.354	0.342	0.329	0.324	0.331
		taud	2.481	2.554	2.532	2.431	2.389	2.445	2.507	2.498	2.526	2.566	2.521	2.403
	Edn at Noon		758	864	901	902	890	900	909	891	872	833	757	694
		Edh at Noon	66	74	88	106	115	109	101	98	87	72	63	64
(44) All-Sky Solar Radiation	RadAvg		1.01	1.87	2.79	4.09	5.18	5.66	6.32	5.41	3.97	2.24	1.19	0.83
	RadStd		0.15	0.26	0.34	0.39	0.43	0.58	0.48	0.38	0.37	0.25	0.16	0.12

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page